

School of Computer Science and Engineering
B.Tech Computer Science and Information Technology

Course Code: B22CI0603	Cloud Computing Lab Project Report	Academic Year: 2025-26
		Semester & Batch: VI
Project Details:		
Project Title:	Evaluating SimGrid and CloudSim Plus for Hybrid Cloud Scientific Workflows	
Place of Project:	REVA UNIVERSITY, BENGALURU	
Student Details:		
Name:	Namitha. NG, Keerthi Shree, Mrudula.G, Nivedha.S	Sign:
Mobile No:	+91 78925 90004, +91 82488 75299, +91 78999 38228, +917368969697	
Email-ID:	ugcet2300357@reva.edu.in , 23020101479@reva.edu.in , 23020105211@reva.edu.in ,	
SRN:	R23EJ082, R23EJ066, R23EJ081, R23EJ089	
Guide and Lab Faculty Members Details		
Guide Name: (The Faculty Member Assigned)	Prof Shilpa V	Sign: Date:
Grade by Guide:		
Name of Lab Co-Faculty 1		Sign: Date:
Grade by Lab Faculty Members (combined)		
SEE Examiners		
Name of Examiner 1:		Sign: Date:
Name of Examiner 2:		Sign: Date:

Contents

1. Title Page	Pg no 1
2. Abstract	Pg no 3
3. Introduction	Pg no 3
4. Literature Review	Pg no 4
5. Methodology	Pg no 5
6. Results and Discussion	Pg no 6
7. Conclusions	Pg no 7
8. References	Pg no 7
9. Appendices	Pg no 8

1. ABSTRACT

This project presents a detailed simulation of a hybrid cloud computing environment using Java and CloudSim Plus to analyze performance and resource utilization. Hybrid cloud combines the advantages of both private and public cloud infrastructures, enabling flexible and cost-effective workload management. In this work, two datacenters are modeled to represent private and public clouds, each consisting of multiple hosts with defined computational resources such as CPU, memory, bandwidth, and storage. Virtual machines are created and deployed on these hosts to execute user workloads in a virtualized environment.

The workload is represented using cloudlets, which simulate real-world computational tasks with varying execution lengths. A broker is responsible for managing the allocation of these tasks to available virtual machines, ensuring efficient execution within the simulated environment. The simulation is performed under different configurations by varying the number of hosts in private and public datacenters, allowing analysis of system behavior under different resource conditions.

The performance of the system is evaluated using key metrics such as makespan, cost analysis, and virtual machine utilization. Makespan represents the total time required to complete all tasks, while cost analysis evaluates the expenses associated with using private and public cloud resources. VM utilization measures how effectively workloads are distributed across available virtual machines. The results are further visualized using a graphical representation to provide a clear understanding of workload distribution and system efficiency.

Overall, this project demonstrates how hybrid cloud environments can be effectively simulated to study performance characteristics without requiring real infrastructure. It highlights the importance of resource allocation and scheduling strategies in improving system efficiency, reducing execution time, and optimizing cost in cloud computing environments.

2. INTRODUCTION

Cloud computing has become an essential technology for delivering scalable and on-demand computing resources over the internet. Among different cloud models, the hybrid cloud approach has gained importance due to its ability to combine the advantages of both private and public clouds.

A hybrid cloud provides better flexibility, improved resource utilization, and cost efficiency by allowing workloads to be distributed across different environments.

In this project, a hybrid cloud environment is simulated using Java and CloudSim Plus. The simulation models multiple datacenters representing private and public cloud infrastructures. Each datacenter consists of hosts that act as physical servers with computing resources such as CPU, memory, and storage.

Virtual machines are created on these hosts to perform computational tasks in a virtualized manner. The workload is represented using cloudlets, which simulate real-world user applications or jobs. A broker is responsible for managing the allocation of cloudlets to virtual machines efficiently. Different resource configurations are tested to analyze system performance under varying conditions.

The project focuses on evaluating important performance metrics such as makespan, cost, and VM utilization. Makespan helps in understanding the total execution time of tasks in the system. Cost analysis provides insight into the economic efficiency of using private and public cloud resources. VM utilization shows how effectively the available resources are being used.

The simulation results are visualized using graphical representation to better understand workload distribution. This project helps in studying how hybrid cloud environments can improve performance and optimize resource usage. It also demonstrates the importance of simulation tools in analyzing complex cloud systems without the need for real infrastructure.

3. LITERATURE REVIEW

Recent studies show that hybrid cloud computing has become an important solution for balancing cost, scalability, and performance in modern computing environments. Research highlights that hybrid cloud integrates private and public cloud infrastructures, allowing organizations to manage sensitive data securely in private clouds while utilizing the scalability of public clouds for high workloads . However, managing such environments is complex due to challenges in resource allocation, scheduling, and cost optimization.

To address these challenges, simulation tools like CloudSim Plus are widely used. Studies indicate that simulation helps in evaluating cloud performance without deploying real infrastructure, which is costly and difficult to manage. CloudSim-based models allow researchers to test different scheduling and allocation strategies, improving resource utilization and reducing execution time . Earlier works also demonstrate that CloudSim provides flexible support for modeling datacenters, virtual machines, and task scheduling policies in a controlled environment.

Recent research in hybrid cloud simulation focuses on workload management and scalability. For example, studies comparing simulation tools such as CloudSim Plus and SimGrid show that they are effective in modeling scientific workflows and analyzing performance in hybrid cloud environments . Other works emphasize the importance of efficient resource allocation techniques, as improper scheduling can lead to increased cost and poor system performance .

Furthermore, modern approaches incorporate optimization techniques and intelligent algorithms to improve load balancing and host utilization in hybrid clouds. Research shows that advanced methods can significantly enhance performance by predicting workload patterns and distributing tasks efficiently across resources . Simulation-based evaluation is therefore essential for testing such techniques before real-world implementation.

Overall, the literature indicates that hybrid cloud environments offer significant advantages but require efficient management strategies. Simulation tools like CloudSim Plus play a crucial role in analyzing system behavior, improving scheduling policies, and optimizing performance metrics such as makespan, cost, and resource utilization. This project builds upon these research contributions by implementing a hybrid cloud simulation and evaluating its performance under different configurations.

4. METHODOLOGY

The methodology of this project is based on simulating a hybrid cloud environment using Java and CloudSim Plus. The simulation begins with the initialization of the CloudSim environment, which provides the platform to model and manage cloud resources. Two datacenters are created to represent the private cloud and public cloud infrastructures. Each datacenter consists of multiple hosts, where each host acts as a physical server with defined processing power, memory, bandwidth, and storage capacity.

Virtual machines are then created and deployed on these hosts to perform computational tasks. A fixed number of virtual machines are configured with specific parameters such as CPU cores, RAM, and bandwidth. The workload is generated in the form of cloudlets, which represent independent user tasks with varying computational lengths. A broker is used to manage the system by assigning cloudlets to the available virtual machines for execution.

The simulation is executed under different scenarios by varying the number of hosts in private and public datacenters, such as small, medium, and high configurations. During execution, the system processes all cloudlets and records their completion details. Performance metrics are then calculated, including makespan, which represents the total execution time, cost analysis based on resource usage in private and public clouds, and VM utilization, which indicates how tasks are distributed across virtual machines.

Finally, the results are analyzed and visualized using a graphical representation to better understand workload distribution among virtual machines. This methodology helps in evaluating the efficiency of the hybrid cloud environment and understanding how resource allocation impacts overall system performance.

ent is designed to replicate a simplified cloud infrastructure consisting of:

- A datacenter representing the computing environment
- A host system managing computational resources
- Multiple virtual machines (VMs) representing subjects
- **Resource Utilization:** Efficiency of time usage
- **Deadline Compliance:** Ability to meet study deadlines

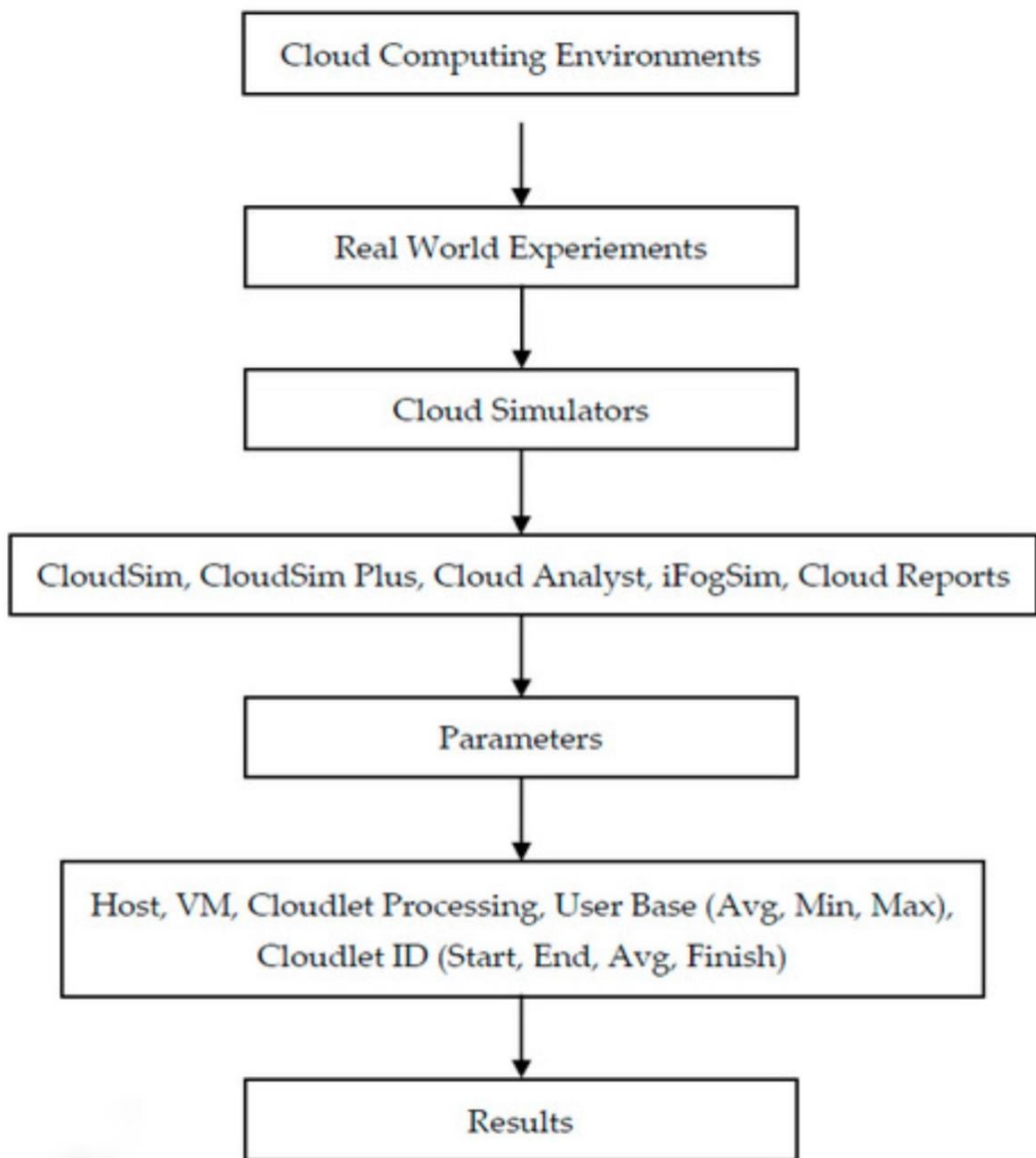
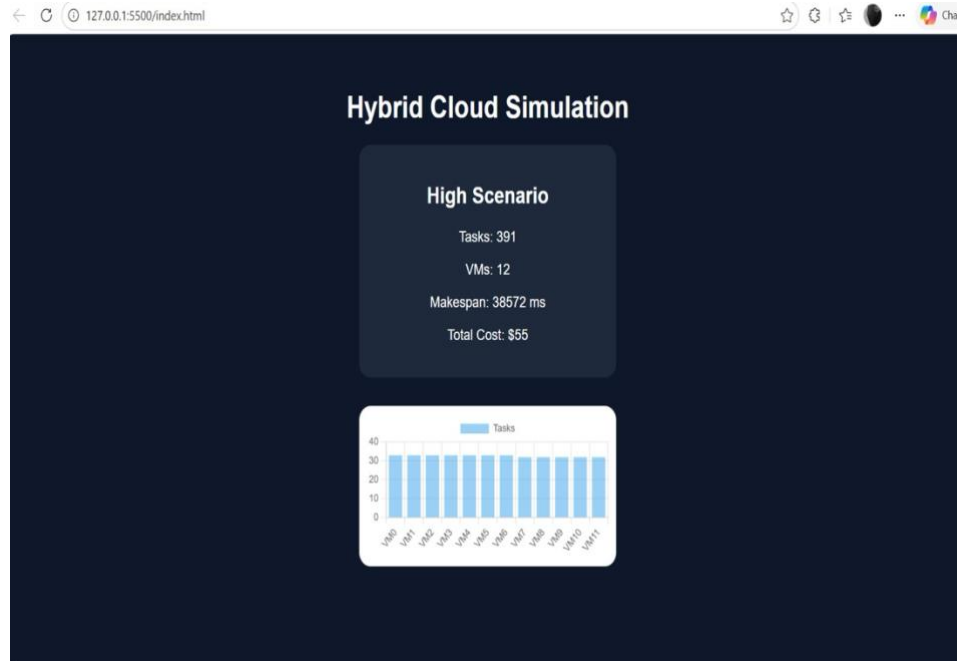


Fig 1: Methodology

5. RESULTS AND DISCUSSION



The simulation was executed under three different configurations representing small, medium, and high hybrid cloud environments using Java and CloudSim Plus. Each scenario varied in the number of private and public hosts, while the number of virtual machines and cloudlets remained constant. The results obtained from the simulation include makespan, cost analysis, and VM utilization, which were used to evaluate system performance.

The makespan increased gradually from the small configuration to the high configuration due to the increase in workload distribution and resource usage. However, the system was able to handle all tasks efficiently in each case without failure. The cost analysis showed that public cloud resources contributed more to the total cost compared to private cloud resources, as public cloud usage was assigned a higher cost factor. This highlights the importance of balancing resource allocation to minimize overall expenses in a hybrid environment.

VM utilization results indicated that the workload was distributed across all virtual machines, but not always equally. Some VMs handled more tasks than others, showing that the default scheduling policy does not guarantee perfect load balancing. This uneven distribution suggests that more advanced scheduling algorithms could further improve performance and efficiency.

The graphical representation of VM utilization provided a clear visualization of workload distribution, making it easier to identify heavily loaded and underutilized virtual machines. Overall, the results demonstrate that the hybrid cloud model is effective in handling large-scale workloads, but performance can be further optimized by improving task scheduling and resource allocation strategies.

Analysis: The analysis of the simulation shows that the hybrid cloud environment implemented using Java and CloudSim Plus is capable of efficiently executing a large number of tasks across multiple configurations. The system performance varies with the number of hosts, indicating that resource availability directly impacts execution time and overall efficiency. Cost evaluation highlights that increased reliance on public cloud resources leads to higher expenses, emphasizing the need for balanced resource usage. VM utilization results reveal uneven task distribution, suggesting limitations in the default scheduling approach. Overall, the analysis indicates that performance can be further improved by adopting optimized scheduling and load balancing techniques.

6. CONCLUSION

This project successfully demonstrates the simulation of a hybrid cloud computing environment using Java and CloudSim Plus. The system effectively models both private and public datacenters to study resource utilization and performance. The implementation shows that virtual machines can handle a large number of cloudlets efficiently under different configurations.

The use of multiple scenarios helps in understanding how changes in infrastructure affect system behavior. The results indicate that the hybrid cloud approach provides flexibility in managing workloads across different environments. Cost analysis highlights the trade-off between using private and public cloud resources.

The study also shows that increasing public cloud usage leads to higher overall cost. VM utilization results reveal that workload distribution is not perfectly balanced among all virtual machines. This suggests the need for improved scheduling strategies for better performance. The graphical representation helps in clearly visualizing workload distribution and system efficiency. Overall, the project proves that hybrid cloud simulation is useful for analyzing performance and optimizing resource allocation in cloud computing systems.

7. REFERENCES

1. CloudSim Plus Documentation, “CloudSim Plus: A modern framework for modeling and simulation of cloud computing environments,” Available at: <https://cloudsimplus.org/>
2. Rajkumar Buyya, J. Broberg, and A. Goscinski, Cloud Computing: Principles and Paradigms, Wiley Publications, 2011. This book explains cloud architecture, virtualization, and resource management techniques.
3. Rajkumar Buyya et al., “CloudSim: A Toolkit for Modeling and Simulation of Cloud Computing Environments and Evaluation of Resource Provisioning Algorithms,” 2009. This paper introduces CloudSim and its role in cloud research.
4. “Efficient Resource Allocation in Hybrid Cloud Environments Using Simulation Techniques,” Research Paper, 2024. This paper discusses workload distribution and cost optimization in hybrid clouds.
5. M. Armbrust et al., “A View of Cloud Computing,” Communications of the ACM, 2010. This paper provides an overview of public, private, and hybrid cloud models.
6. “Performance Evaluation of Cloud Computing Systems Using Simulation Tools,” Springer Journal Article. It highlights the importance of simulation tools for analyzing makespan, cost, and VM utilization.

